SSIE 553 – Homework 5

Linear Programming Duality

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Course: SSIE 553 – Operations Research

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This homework covers LP duality theory: constructing dual problems, graphical solutions, weak duality, complementary slackness, and shadow price interpretation.

Background: LP Duality Rules

Every Linear Program (primal) has a corresponding dual problem. The duality relationship is symmetric: the dual of the dual is the primal. Below are the complete transformation rules used throughout this homework.

Rules for MAX Primal → MIN Dual:

Primal (MAX): Max $c^T x$, s.t. $Ax \leq b$ (or $\geq$ or $=$), $x \geq 0$

Row → Dual variable correspondence:

- Primal constraint $\leq$ → Dual variable $y_i \geq 0$
- Primal constraint $\geq$ → Dual variable $y_i \leq 0$
- Primal constraint $=$ → Dual variable y_i unrestricted (urs)

Column → Dual constraint correspondence:

- Primal $x_j \geq 0$ → Dual constraint j: (col j of $A)^T y \geq c_j$
- Primal x_j urs → Dual constraint j: (col j of $A)^T y = c_j$

Dual Objective: MIN $b^T y$

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Dual Objective: MAX $b^T x$

Key Theorem – Strong Duality:

If the primal has an optimal solution x^*, the dual also has an optimal y^*, and their objectives are equal: $c^S x^* = b^S y^*$.

Weak Duality: For any primal feasible x and dual feasible y:

$c^S x \leq b^S y$ (MAX primal / MIN dual)

Question 1: Determine the Dual of Each LP

Part (i) [10 pts]

PRIMAL (MAX):

Max $z = 2x_1 + x_2$

s.t.

$-x_1 + x_2 \leq 1$ ...(1)

$x_1 + x_2 \leq 3$ ...(2)

$x_1 - 2x_2 \leq 4$ ...(3)

$x_1, x_2 \geq 0$

Step 1 – Assign dual variables (each ≤ constraint → y_i ≥ 0):

Constraint (1) is ≤ → $y_1 \geq 0$

Constraint (2) is ≤ → $y_2 \geq 0$

Constraint (3) is ≤ → $y_3 \geq 0$

Step 2 – Dual objective (MIN; RHS values become obj. coefficients):

MIN $w = 1 \cdot y_1 + 3 \cdot y_2 + 4 \cdot y_3$

Step 3 – Dual constraints (each x_j ≥ 0 → constraint ≥ c_j):

For x_1 (≥ 0): read coefficients of x_1 down each row: (-1), (+1), (+1)

⇒ $-y_1 + y_2 + y_3 \geq 2$ [$c_1 = 2$]

For x_2 (≥ 0): coefficients of x_2 in rows: (+1), (+1), (-2)

⇒ $y_1 + y_2 - 2y_3 \geq 1$ [$c_2 = 1$]

DUAL (Part i):

Min $w = y_1 + 3y_2 + 4y_3$

s.t.

$-y_1 + y_2 + y_3 \geq 2$

$y_1 + y_2 - 2y_3 \geq 1$

$y_1, y_2, y_3 \geq 0$

Part (ii) [10 pts]

PRIMAL (MIN):

Min $w = y_1 - y_2$

s.t.

$2y_1 + y_2 \geq 4$ ...(5)

$y_1 + y_2 \geq 1$ ...(6)

$y_1 + 2y_2 \geq 3$ ...(7)

$y_1, y_2 \geq 0$

Step 1 – Assign dual variables (MIN primal → MAX dual; ≥ → x_i ≥ 0):

Constraint (5) is ≥ → $x_1 \geq 0$

Constraint (6) is ≥ → $x_2 \geq 0$

Constraint (7) is ≥ → $x_3 \geq 0$

Step 2 – Dual objective (MAX; RHS becomes obj. coeff.):

MAX $z = 4x_1 + 1 \cdot x_2 + 3x_3$

Step 3 – Dual constraints (each y_j ≥ 0 → constraint ≤ c_j):

For y_1 (≥ 0): coefficients in rows: 2, 1, 1

⇒ $2x_1 + x_2 + x_3 \leq 1$ [$c_1 = 1$]

For y_2 (≥ 0): coefficients in rows: 1, 1, 2

⇒ $x_1 + x_2 + 2x_3 \leq -1$ [$c_2 = -1$]

DUAL (Part ii):

Max $z = 4x_1 + x_2 + 3x_3$

s.t.

$2x_1 + x_2 + x_3 \leq 1$

$x_1 + x_2 + 2x_3 \leq -1$

$x_1, x_2, x_3 \geq 0$

Part (iii) [10 pts]

PRIMAL (MAX) – mixed constraints and unrestricted variables:

Max $z = 4x_1 - x_2 + 2x_3$

s.t.

$x_1 + x_2 \le 5$ ...(9)

$2x_1 + x_2 \le 7$ ...(10)

$2x_2 + x_3 \ge 6$ ...(11)

$x_1 + x_3 = 4$ ...(12)

$x_1 \ge 0$; x_2, x_3 unrestricted (urs)

Step 1 – Assign dual variables:

Constraint (9) is $\le$ → $y_1 \ge 0$

Constraint (10) is $\le$ → $y_2 \ge 0$

Constraint (11) is $\ge$ → $y_3 \le 0$

Constraint (12) is $=$ → y_4 unrestricted (urs)

Step 2 – Dual objective:

MIN $w = 5y_1 + 7y_2 + 6y_3 + 4y_4$

Step 3 – Dual constraints:

For x_1 (≥ 0) → constraint $\ge c_1 = 4$:

Coefficients in (9),(10),(11),(12): 1, 2, 0, 1

⇒ $y_1 + 2y_2 + 0 \cdot y_3 + y_4 \ge 4$

For x_2 (urs) → constraint $= c_2 = -1$:

Coefficients in (9),(10),(11),(12): 1, 1, 2, 0

⇒ $y_1 + y_2 + 2y_3 = -1$

For x_3 (urs) → constraint $= c_3 = 2$:

Coefficients in (9),(10),(11),(12): 0, 0, 1, 1

⇒ $y_3 + y_4 = 2$

DUAL (Part iii):

Min $w = 5y_1 + 7y_2 + 6y_3 + 4y_4$

s.t.

$y_1 + 2y_2 + y_4 \ge 4$

$y_1 + y_2 + 2y_3 = -1$

$y_3 + y_4 = 2$

$y_1, y_2 \ge 0$; $y_3 \le 0$; y_4 urs

Part (iv) [10 pts]

PRIMAL (MIN) – mixed constraints and unrestricted variables:

Min $w = 4y_1 + 2y_2 - y_3$

s.t.

$y_1 + 2y_2 \leq 6$ ...(14)

$y_1 - y_2 + 2y_3 = 8$ ...(15)

$y_1 \geq 0$; y_2, y_3 unrestricted (urs)

Step 1 – Assign dual variables (MIN → MAX dual):

Constraint (14) is $\leq$ → $x_1 \leq 0$

Constraint (15) is $=$ → x_2 unrestricted (urs)

Step 2 – Dual objective:

MAX $z = 6x_1 + 8x_2$

Step 3 – Dual constraints:

For y_1 (≥ 0) → constraint $\leq c_1 = 4$:

Coefficients in (14),(15): 1, 1

$\Rightarrow x_1 + x_2 \leq 4$

For y_2 (urs) → constraint $= c_2 = 2$:

Coefficients in (14),(15): 2, -1

$\Rightarrow 2x_1 - x_2 = 2$

For y_3 (urs) → constraint $= c_3 = -1$:

Coefficients in (14),(15): 0, 2

$\Rightarrow 2x_2 = -1$

DUAL (Part iv):

Max $z = 6x_1 + 8x_2$

s.t.

$$x_1 + x_2 \leq 4$$

$$2x_1 - x_2 = 2$$

$$2x_2 = -1$$

$$x_1 \leq 0;\ x_2 \text{ urs}$$

Question 2: Dual Construction + Graphical Solution [10 pts]

Part (i) – Construct the Dual

PRIMAL (MAX):

Max $z = 2x_1 + 3x_2 + 6x_3$

s.t.

$x_1 \quad + x_3 \le 3$ ...(19)

$x_2 + x_3 \le 5$ ...(20)

$x_1, x_2, x_3 \ge 0$

Step 1 – Assign dual variables:

Constraint (19) is $\le$ → $y_1 \ge 0$

Constraint (20) is $\le$ → $y_2 \ge 0$

Step 2 – Dual objective (MIN):

MIN $w = 3y_1 + 5y_2$

Step 3 – Dual constraints (x_j ≥ 0 → ≥ c_j):

Column x_1: coefficients (19)→1, (20)→0 ⇒ $y_1 \ge 2$

Column x_2: coefficients (19)→0, (20)→1 ⇒ $y_2 \ge 3$

Column x_3: coefficients (19)→1, (20)→1 ⇒ $y_1 + y_2 \ge 6$

DUAL:

Min $w = 3y_1 + 5y_2$

s.t.

$y_1 \quad \ge 2$

$y_2 \ge 3$

$y_1 + y_2 \ge 6$

$y_1, y_2 \ge 0$

Part (ii) – Solve the Dual Graphically

Step 1 – Identify constraint boundaries in the y_1-y_2 plane:

L1: $y_1 = 2$ (vertical line; feasible region to the right)

L2: $y_2 = 3$ (horizontal line; feasible region above)

L3: $y_1 + y_2 = 6$ (diagonal line; feasible region above)

Note: if $y_1 \ge 2$ and $y_2 \ge 3$, then $y_1+y_2 \ge 5$, but we need ≥ 6.

So L3 is NOT redundant; it cuts off the corner near (2, 3).

Step 2 – Find corner (extreme) points of the feasible region:

Corner A: intersection of L1 and L3

$y_1 = 2$ and $y_1 + y_2 = 6$ ⇒ $y_2 = 4$

$\Rightarrow$ Corner A = (2, 4)

Corner B: intersection of L2 and L3

$y_2 = 3$ and $y_1 + y_2 = 6 \Rightarrow y_1 = 3$

$\Rightarrow$ Corner B = (3, 3)

The feasible region extends to the upper-right beyond these corners

(the objective increases in that direction, so the minimum is at a corner).

Step 3 – Evaluate dual objective at each corner:

Objective: $w = 3y_1 + 5y_2$

Corner A = (2, 4): $w = 3(2) + 5(4) = 6 + 20 = 26$

Corner B = (3, 3): $w = 3(3) + 5(3) = 9 + 15 = 24$

Minimum w = 24 at Corner B = $(y_1^*, y_2^*) = (3, 3)$.

DUAL OPTIMAL SOLUTION:

$y_1^* = 3$, $y_2^* = 3$, $w^* = 24$

SHADOW PRICES (dual variables = shadow prices of primal constraints):

Shadow price of constraint (19) [$x_1 + x_3 \le 3$]: $\pi_1 = y_1^* = 3$

Shadow price of constraint (20) [$x_2 + x_3 \le 5$]: $\pi_2 = y_2^* = 3$

Interpretation: Each additional unit of RHS in constraint (19) or (20) increases the primal optimal profit by 3 units.

Verification via Strong Duality:

Primal solution: $x_1=0$, $x_2=2$, $x_3=3$ → $z = 0+6+18 = 24 = w^*$ \[OK]

Question 3: Infeasible Primal / Unbounded Dual [15 pts]

Part (i) – Primal Has No Feasible Solution (Graphical Proof)

PRIMAL (MAX):

Max $z = x_1 + 2x_2$

s.t.

$-x_1 + x_2 \le -2$ ...(24) equivalently: $x_2 \le x_1 - 2$

$4x_1 + x_2 \le 4$ ...(25) equivalently: $x_2 \le 4 - 4x_1$

$x_1, x_2 \ge 0$

Step 1 – Derive implied lower bound on x_1:

From constraint (24): $x_2 \le x_1 - 2$

From non-negativity: $x_2 \ge 0$

Combining: $0 \le x_2 \le x_1 - 2$

This requires: $x_1 - 2 \ge 0 \Rightarrow x_1 \ge 2$

Step 2 – Substitute $x_1 \ge 2$ into constraint (25):

From (25): $x_2 \le 4 - 4x_1$

If $x_1 \ge 2$: $4 - 4x_1 \le 4 - 4(2) = 4 - 8 = -4$

So: $x_2 \le -4 < 0$

But $x_2 \ge 0$ is required. This is a CONTRADICTION.

Step 3 – Graphical interpretation:

On the y-axis (x_2 axis):

Region from (24) with $x_2 \ge 0$: starts at $x_1=2$, allowing x_2 between 0 and x_1-2.

Region from (25): for $x_1 \ge 2$, x_2 must be $\le 4-4x_1 \le -4$ (below x-axis).

These two half-planes do not intersect in the first quadrant ($x_1, x_2 \ge 0$).

CONCLUSION (Part i):

The primal feasible region is EMPTY. No (x_1, x_2) with $x_1, x_2 \ge 0$ can simultaneously satisfy constraints (24) and (25).

The primal problem has NO FEASIBLE SOLUTION.

Part (ii) – Construct the Dual

Step 1 – Assign dual variables:

Constraint (24) is $\le$ → $y_1 \ge 0$

Constraint (25) is $\le$ → $y_2 \ge 0$

Step 2 – Dual objective:

MIN $w = -2y_1 + 4y_2$

Step 3 – Dual constraints:

Column x_1: coeff in (24)→-1, (25)→4 ⇒ $-y_1 + 4y_2 \geq 1$

Column x_2: coeff in (24)→1, (25)→1 ⇒ $y_1 + y_2 \geq 2$

DUAL:

Min $w = -2y_1 + 4y_2$

s.t.

$-y_1 + 4y_2 \geq 1$

$y_1 + y_2 \geq 2$

$y_1, y_2 \geq 0$

Part (iii) – Dual is Unbounded (Graphical Proof)

Step 1 – Confirm the dual is feasible:

Try $(y_1, y_2) = (1, 1)$:

Constraint (i): $-1 + 4(1) = 3 \geq 1$ \[OK]

Constraint (ii): $1 + 1 = 2 \geq 2$ \[OK]

$y_1, y_2 \geq 0$ \[OK]

$w = -2(1) + 4(1) = 2$

The dual is feasible.

Step 2 – Find a ray along which w → -∞:

Let $y_1 \to +\infty$ and set $y_2 = (1 + y_1)/4$ (minimum from constraint i)

Objective along this ray:

$w = -2y_1 + 4 \times (1+y_1)/4 = -2y_1 + 1 + y_1 = -y_1 + 1$

As $y_1 \to +\infty$: $w \to -\infty$

Check constraint (ii) along this ray:

$y_1 + y_2 = y_1 + (1+y_1)/4 = (5y_1+1)/4 \to +\infty \geq 2$ \[OK]

Check $y_2 \geq 0$: $y_2 = (1+y_1)/4 > 0$ \[OK]

Step 3 – Graphical interpretation:

On the y_1-y_2 plane, the feasible region is unbounded in the direction of increasing y_1 (to the right). The objective contours $w = -2y_1 + 4y_2$ have gradient (-2, 4), meaning the objective DECREASES as y_1 increases (for fixed w, moving right reduces w). Since we can move $y_1 \to \infty$ while staying feasible, $w \to -\infty$ and the problem is UNBOUNDED.

CONCLUSION (Part iii):

The dual is UNBOUNDED ($w \to -\infty$ along the ray $y_1 \to \infty$, $y_2 = (1+y_1)/4$).

This is consistent with LP Duality Theory:

Primal INFEASIBLE ⇒ Dual is either INFEASIBLE or UNBOUNDED.

Here the dual is feasible but unbounded, confirming the primal has no feasible solution.

Question 4: Weak Duality and Complementary Slackness [20 pts]

PRIMAL (MAX):

Max $z = 2x_1 + 5x_2 + 4x_3$

s.t.

$x_1 + 2x_2 + 4x_3 \le 10$ ...(29)

$3x_1 + 3x_2 + 2x_3 \le 10$ ...(30)

$x_1, x_2, x_3 \ge 0$

Part (i) – Construct the Dual

Step 1 – Assign dual variables:

Constraint (29) is ≤ → $y_1 \ge 0$

Constraint (30) is ≤ → $y_2 \ge 0$

Step 2 – Dual objective:

MIN $w = 10y_1 + 10y_2$

Step 3 – Dual constraints:

Column x_1: coeff (29)→1, (30)→3 ⇒ $y_1 + 3y_2 \ge 2$

Column x_2: coeff (29)→2, (30)→3 ⇒ $2y_1 + 3y_2 \ge 5$

Column x_3: coeff (29)→4, (30)→2 ⇒ $4y_1 + 2y_2 \ge 4$

DUAL:

Min $w = 10y_1 + 10y_2$

s.t.

$y_1 + 3y_2 \ge 2$

$2y_1 + 3y_2 \ge 5$

$4y_1 + 2y_2 \ge 4$

$y_1, y_2 \ge 0$

Part (ii) – Weak Duality: Optimal Primal Value ≤ 25

Weak Duality Theorem:

For ANY primal feasible x and ANY dual feasible y:

$z = c^T x \le b^T y = w$

Therefore: optimal $z^* \le$ (value of any feasible dual solution).

We find a dual feasible solution with $w = 25$.

Find a feasible dual solution:

Try $y_1 = 5/2 = 2.5$, $y_2 = 0$:

Constraint 1: $y_1 + 3y_2 = 2.5 + 0 = 2.5 \ge 2$ \[OK]

Constraint 2: $2y_1 + 3y_2 = 5.0 + 0 = 5.0 \geq 5$ \[OK]

Constraint 3: $4y_1 + 2y_2 = 10.0 + 0 = 10 \geq 4$ \[OK]

$y_1, y_2 \geq 0$: $2.5 \geq 0, 0 \geq 0$ \[OK]

Dual objective: $w = 10(2.5) + 10(0) = 25$

CONCLUSION (Part ii):

$(y_1, y_2) = (2.5, 0)$ is dual feasible with $w = 25$.

By Weak Duality: $z^* \leq w = 25$

⇒ The optimal primal objective value cannot exceed 25. \[OK]

Part (iii) – Basic Solution (x_2, x_3 basic) + Complementary Slackness

Step 1 – Set up basis matrix B = [A_2 | A_3]:

Non-basic variable: $x_1 = 0$.

Basis columns from constraint matrix (rows = constraints (29),(30)):

A_2 = column of x_2 = $[2, 3]^T$

A_3 = column of x_3 = $[4, 2]^T$

$$B = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$$

$\det(B) = (2)(2) - (4)(3) = 4 - 12 = -8$

Step 2 – Compute B^{-1}:

For 2×2 matrix [[a,b],[c,d]]: $B^{-1} = (1/\det) \times [[d,-b],[-c,a]]$

$B^{-1} = (1/-8) \times [[2, -4], [-3, 2]]$

$= [[-1/4, 1/2], [3/8, -1/4]]$

Step 3 – Compute basic variable values $x_B = B^{-1}b$ ($b = [10, 10]^S$):

$x_2 = (-1/4)(10) + (1/2)(10) = -2.5 + 5.0 = 2.5$

$x_3 = (3/8)(10) + (-1/4)(10) = 3.75 - 2.5 = 1.25$

Full primal solution: $x_1 = 0$, $x_2 = 2.5$, $x_3 = 1.25$

Primal objective:

$z = 2(0) + 5(2.5) + 4(1.25) = 0 + 12.5 + 5.0 = 17.5$

Step 4 – Derive dual solution: $y^* = c\_B \times B^{-1}$:

$c\_B = [c_2, c_3] = [5, 4]$ (objective coefficients of basic variables)

$y_1^* = c\_B \cdot (\text{col 1 of } B^{-1})$
$= 5(-1/4) + 4(3/8)$
$= -5/4 + 12/8 = -10/8 + 12/8 = 2/8 = 0.25$

$y_2^* = c\_B \cdot (\text{col 2 of } B^{-1})$
$= 5(1/2) + 4(-1/4)$
$= 5/2 - 1 = 3/2 = 1.5$

Dual solution: $y_1^* = 0.25$, $y_2^* = 1.5$

Dual objective: $w = 10(0.25) + 10(1.5) = 2.5 + 15 = 17.5$

Step 5 – Verify Complementary Slackness (CS):

CS Condition A: $y\_i^* \times$ (primal slack of constraint i) = 0 for all i

Constraint (29): $x_1+2x_2+4x_3 = 0 + 5 + 5 = 10 \Rightarrow$ slack $= 10-10 = 0$
$y_1^* = 0.25 > 0$; slack $= 0 \Rightarrow 0.25 \times 0 = 0$ \[OK]

Constraint (30): $3x_1+3x_2+2x_3 = 0 + 7.5 + 2.5 = 10 \Rightarrow$ slack $= 0$
$y_2^* = 1.5 > 0$; slack $= 0 \Rightarrow 1.5 \times 0 = 0$ \[OK]

CS Condition B: $x\_j^* \times$ (dual slack of constraint j) = 0 for all j

For x_1: dual constraint is $y_1+3y_2 \geq 2$
LHS $= 0.25 + 4.5 = 4.75 \Rightarrow$ dual slack $= 4.75-2 = 2.75$
$x_1^* = 0 \Rightarrow 0 \times 2.75 = 0$ \[OK]

For x_2: dual constraint is $2y_1+3y_2 \geq 5$
LHS $= 0.5 + 4.5 = 5.0 \Rightarrow$ dual slack $= 0$
$x_2^* = 2.5 > 0$; dual slack $= 0 \Rightarrow 2.5 \times 0 = 0$ \[OK]

For x_3: dual constraint is $4y_1+2y_2 \geq 4$
LHS $= 1.0 + 3.0 = 4.0 \Rightarrow$ dual slack $= 0$
$x_3^* = 1.25 > 0$; dual slack $= 0 \Rightarrow 1.25 \times 0 = 0$ \[OK]

CONCLUSION (Part iii):

Primal: $x_1=0$, $x_2=2.5$, $x_3=1.25 \Rightarrow z = 17.5$

Dual: $y_1=0.25$, $y_2=1.5$ ⇒ $w = 17.5$

Since $z = w = 17.5$ (objectives are equal), and all Complementary Slackness conditions are satisfied, BOTH solutions are OPTIMAL.

The true optimal is 17.5 (within the bound of ≤ 25 shown in Part ii).

Question 5: Shadow Prices and Willingness-to-Pay [15 pts]

LP:

Max $z = 3x_1 + 7x_2 + 5x_3$

s.t.

$x_1 + x_2 + x_3 \le 50$ ... Sugar Capacity

$2x_1 + 3x_2 + x_3 \le 100$ ... Chocolate Capacity

$x_1, x_2, x_3 \ge 0$

Optimal basis: $\{x_3, x_2\}$ (ordered: x_3 first, x_2 second)

$$B^{-1} = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

Part (i) – Determine Shadow Prices

Step 1 – Identify basis columns and verify B^{-1}:

The basis is ordered $\{x_3, x_2\}$.

From the constraint matrix (rows = Sugar, Chocolate constraints):

$A_3 = [1, 1]^T$ (x_3 has coefficient 1 in both constraints)

$A_2 = [1, 3]^T$ (x_2 has coefficient 1 in sugar, 3 in chocolate)

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$$

$\det(B) = (1)(3) - (1)(1) = 3 - 1 = 2$

$B^{-1} = (1/2)\times[[3, -1],[-1, 1]] = [[3/2, -1/2],[-1/2, 1/2]]$

This matches the given B^{-1}. \[OK]

Step 2 – Compute shadow prices: $y^* = c_B \times B^{-1}$:

$c_B = [c_3, c_2] = [5, 7]$ (objective coefficients in basis order)

$y_1^* = c_B \cdot (\text{column 1 of } B^{-1})$ [shadow price of Sugar]

$= 5(3/2) + 7(-1/2)$

$= 15/2 - 7/2 = 8/2 = 4$

$y_2^* = c_B \cdot (\text{column 2 of } B^{-1})$ [shadow price of Chocolate]

$= 5(-1/2) + 7(1/2)$

$= -5/2 + 7/2 = 2/2 = 1$

SHADOW PRICES:

$y_1^* = 4$ cents per oz (Sugar Capacity constraint)

$y_2^* = 1$ cent per oz (Chocolate Capacity constraint)

Interpretation:

Sugar shadow price = 4: Each additional 1 oz of sugar (beyond the current 50 oz limit) allows Company X to increase its total profit by 4 cents, at the current optimal basis.

Chocolate shadow price = 1: Each additional 1 oz of chocolate (beyond the current 100 oz limit) increases total profit by 1 cent at the current optimal basis.

Sugar is the more valuable (binding) resource in this solution, as it has a higher marginal value (4 > 1).

Part (ii) – Maximum Willingness-to-Pay for 1 Extra oz of Sugar

Reasoning using shadow prices:

The shadow price $y_1^* = 4$ is the marginal value of 1 additional oz of sugar.

If Company X pays a price P per extra oz of sugar, the net benefit is:

Net benefit = marginal value − cost = $4 - P$

Company X will purchase extra sugar as long as it does not lose money:

$4 - P \geq 0 \Rightarrow P \leq 4$

The break-even point is at $P = 4$. Paying more than 4 cents per oz would reduce the company's net profit.

ANSWER (Part ii):

Maximum willingness-to-pay for 1 extra oz of Sugar = 4 cents

Company X should not pay more than 4 cents per additional oz of sugar.

Part (iii) – Maximum Willingness-to-Pay for 1 Extra oz of Chocolate

Reasoning using shadow prices:

The shadow price $y_2^* = 1$ is the marginal value of 1 additional oz of chocolate.

If Company X pays a price P per extra oz of chocolate, the net benefit is:

Net benefit = $1 - P$

Company X will purchase extra chocolate as long as:

$1 - P \geq 0 \Rightarrow P \leq 1$

ANSWER (Part iii):

Maximum willingness-to-pay for 1 extra oz of Chocolate = 1 cent

Company X should not pay more than 1 cent per additional oz of chocolate.

Chocolate is less valuable at the margin than sugar in this solution.

Summary of All Results

Q1(i)	**Dual of MAX $2x_1+x_2$**	Min $y_1+3y_2+4y_3$ s.t. $-y_1+y_2+y_3\geq 2$, $y_1+y_2-2y_3\geq 1$
Q1(ii)	**Dual of MIN y_1-y_2**	Max $4x_1+x_2+3x_3$ s.t. $2x_1+x_2+x_3\leq 1$, $x_1+x_2+2x_3\leq -1$
Q1(iii)	**Dual of MAX $4x_1-x_2+2x_3$**	Min $5y_1+7y_2+6y_3+4y_4$ (see full solution for constraints)
Q1(iv)	**Dual of MIN $4y_1+2y_2-y_3$**	Max $6x_1+8x_2$ s.t. $x_1+x_2\leq 4$, $2x_1-x_2=2$, $2x_2=-1$
Q2(i)	**Dual of Q2 MAX primal**	Min $3y_1+5y_2$ s.t. $y_1\geq 2$, $y_2\geq 3$, $y_1+y_2\geq 6$
Q2(ii)	**Graphical dual optimal**	$y_1^*=3$, $y_2^*=3$, $w^*=24$; shadow prices $\pi_1=3$, $\pi_2=3$
Q3(i)	**Primal infeasibility**	INFEASIBLE: $x_1\geq 2$ required, but then $x_2\leq -4<0$. Empty region.
Q3(ii)	**Dual of Q3 MAX**	Min $-2y_1+4y_2$ s.t. $-y_1+4y_2\geq 1$, $y_1+y_2\geq 2$
Q3(iii)	**Dual unbounded**	Ray $y_1\to\infty$, $y_2=(1+y_1)/4$ is feasible and gives $w\to-\infty$. UNBOUNDED.
Q4(i)	**Dual of Q4 MAX**	Min $10y_1+10y_2$ s.t. $y_1+3y_2\geq 2$, $2y_1+3y_2\geq 5$, $4y_1+2y_2\geq 4$
Q4(ii)	**Weak duality proof**	$(y_1,y_2)=(2.5,0)$ feasible, $w=25$ ⇒ optimal primal ≤ 25
Q4(iii)	**CS & optimality**	$x^*=(0,2.5,1.25)$ $z=17.5$; $y^*=(0.25,1.5)$ $w=17.5$. $z=w$ ⇒ BOTH OPTIMAL
Q5(i)	**Shadow prices**	$y_1^*=4$ cents/oz (Sugar), $y_2^*=1$ cent/oz (Chocolate)
Q5(ii)	**WTP for Sugar**	Maximum willingness-to-pay = 4 cents per oz
Q5(iii)	**WTP for Chocolate**	Maximum willingness-to-pay = 1 cent per oz

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